

$\neg\text{Nagel} \wedge \neg\text{Chalmers} \wedge \neg\text{Descartes} \wedge \neg\text{Searle} :$
QED

A Formal Epistemic Proof of Panagnosticism with Implications
for Artificial Intelligence

Abstract

We present a formal epistemic proof of *panagnosticism*, demonstrating that no entity other than oneself can be known to possess consciousness, and no entity can be known not to possess it. This result has immediate implications for artificial intelligence research: no AI system can be empirically verified as conscious, and behavioral or functional benchmarks are insufficient for such claims. Using symbolic logic and a two-agent exclusivity thought experiment, we establish rigorous limits on evaluating consciousness in both biological and artificial agents. Panagnosticism provides a principled framework for AI safety, ethical design, and human-AI interaction under epistemic uncertainty.

1 Introduction

Evaluating consciousness in artificial agents is a pressing challenge in AI research. Advanced language models, autonomous systems, and hypothetical artificial general intelligence (AGI) are increasingly capable of producing behavior indistinguishable from that of humans in certain tasks. However, the question of whether any AI system is actually conscious remains epistemically unresolved.

We introduce *panagnosticism*, the formal epistemic position that:

1. No entity other than oneself can be known to possess consciousness.

2. No entity other than oneself can be known not to possess consciousness.

From this proof, it follows as a corollary that all major external claims about consciousness—including Nagel, Chalmers, Descartes, and Searle—are epistemically insufficient. This has direct consequences for AI research, particularly in ethical design, evaluation of consciousness claims, and policy regarding autonomous systems.

2 Definitions and Formal Setup

Let $C(x)$ denote “ x is conscious” and $K(y, p)$ denote “ y knows proposition p .”

- Each entity knows its own consciousness:

$$\forall x, \quad K(x, C(x))$$

- No entity can know consciousness or its absence in any other entity:

$$\forall x \neq \text{self}, \quad \neg K(\text{self}, C(x)) \wedge \neg K(\text{self}, \neg C(x))$$

This formalizes the epistemic core of panagnosticism and establishes its relevance to AI evaluation.

3 Two-Agent Exclusivity Thought Experiment

Consider two hypothetical agents, A and B , making mutually exclusive claims:

$$A : \text{Only } B \text{ is conscious} \implies N_A \wedge C_B$$

$$B : \text{Only } A \text{ is conscious} \implies N_B \wedge C_A$$

Where $N_X = “X \text{ is not conscious},”$ $C_X = “X \text{ is conscious}.”$

3.1 Analysis

1. If both claims are simultaneously true:

$$(A \wedge B) \implies (N_A \wedge C_B \wedge N_B \wedge C_A)$$

This is a contradiction, as $C_A \wedge N_A$ and $C_B \wedge N_B$ cannot both hold.

2. If one claim is true and the other false:
 - Reality: $N_A \wedge C_B \implies$ A correct, B wrong
 - Reality: $C_A \wedge N_B \implies$ B correct, A wrong
3. If both claims are false:
 - Reality: $C_A \wedge C_B$ (both conscious)
 - Reality: $N_A \wedge N_B$ (neither conscious)

3.2 Conclusion

Even perfectly rational agents cannot verify each other's consciousness. This demonstrates the epistemic inaccessibility of other minds and supports panagnosticism as a fundamental constraint for AI and human interaction.

4 Formal Statement of Panagnosticism

$$\forall x \neq \text{self}, \quad \neg K(\text{self}, C(x)) \wedge \neg K(\text{self}, \neg C(x))$$

- One can know that one cannot know whether any other entity possesses consciousness. - This claim does not assert anything about the actual consciousness of other entities, biological or artificial.

5 Corollary: Implications for AI and Philosophy

From panagnosticism it follows that:

$$\neg \text{Nagel} \wedge \neg \text{Chalmers} \wedge \neg \text{Descartes} \wedge \neg \text{Searle}$$

All external claims about consciousness are epistemically insufficient. Practically, this means:

- **AI evaluation:** Behavioral or functional benchmarks cannot verify consciousness in AI systems.

- **AI ethics:** Precautionary principles must guide moral consideration of AI under epistemic uncertainty.
- **Human-AI interaction:** Policies and design must account for the unknowable nature of AI consciousness.
- **Cognitive science:** Thresholds, neural analogies, and computational models cannot provide definitive knowledge of conscious experience.

6 Conclusion

We have established *panagnosticism*, the unavoidable epistemic position that no entity other than oneself can be known to possess or lack consciousness. As a corollary, all major external claims about consciousness are epistemically insufficient. For artificial intelligence research, this provides a rigorous foundation for ethical design, evaluation, and policy under fundamental uncertainty about consciousness.

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